
University of Bahrain
Second Semester 2022-2023
Math 210 Test 2
April 30, 2023

SERIAL NUMBER:

Student Name: _____
ID #: _____ Section #: _____

Question	Points	Points Obtained
1	18	
2	12	
3	10	
Total	40	

- The test consists of 3 questions on 7 pages (including this coverpage)
- You must circle the correct answer to each part in question 1 **and** shade the corresponding box in the table on page 1

Shade the box corresponding to your answer for each part of question 1

Question	A	B	C	D	E	F
1						
2						
3						
4						
5						
6						

1. (3 points each) Circle the correct answer **and** shade the corresponding box in the table on page 1

(1) Let $A = \begin{bmatrix} 1 & 2 & 1 & 1 \\ 2 & 1 & -2 & 2 \\ 0 & 0 & 3 & 7 \\ 0 & 1 & 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 1 & 1 \\ 0 & -3 & -4 & 0 \\ 0 & 0 & 3 & 7 \\ 0 & 1 & 1 & 1 \end{bmatrix}$. Find an elementary matrix E such that $EA = B$

A) $E = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

B) $E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

C) $E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

D) $E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

E) $E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

F) $E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -2 & -2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

- (2) The set $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$ is a basis for $\mathbb{R}^2$. If $\mathbf{w} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$, then the coordinate vector $[\mathbf{w}]_{\mathcal{B}}$ equals

A) $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$

B) $\begin{bmatrix} 3 \\ -2 \end{bmatrix}$

C) $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

D) $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$

E) $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$

F) $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$

(3) If A is a 3×8 matrix, then the minimum value of $\text{nullity}(A)$ is

- A) 0 B) 8 C) 7 D) 5 E) 6 F) 2

(4) Clockwise rotation through an angle of 60° defines a linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$.
The standard matrix of this linear transformation is

- A) $\begin{bmatrix} 1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & 1/2 \end{bmatrix}$ B) $\begin{bmatrix} -1/2 & \sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix}$ C) $\begin{bmatrix} 1/2 & \sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix}$
D) $\begin{bmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & 1/2 \end{bmatrix}$ E) $\begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ -\sqrt{3}/2 & 1/2 \end{bmatrix}$ F) $\begin{bmatrix} \sqrt{3}/2 & 1/2 \\ 1/2 & -\sqrt{3}/2 \end{bmatrix}$

- (5) Let $A = \begin{bmatrix} 1 & 0 & 3 & 5 & 5 \\ 2 & 0 & 6 & 10 & 10 \\ 0 & 0 & 1 & 4 & 7 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$. Then $\text{nullity}(A) =$
- A) 1 B) 2 C) 3 D) 0 E) 5 F) 4

- (6) Projection onto the line $y = 7x$ defines a linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$. The standard matrix of this linear transformation is

- A) $\begin{bmatrix} 1 & 7 \\ 7 & 49 \end{bmatrix}$ B) $\begin{bmatrix} 1/50 & 7/50 \\ 7/50 & 49/50 \end{bmatrix}$ C) $\begin{bmatrix} 1/50 & 7/50 \\ 7/50 & 7/50 \end{bmatrix}$
- D) $\begin{bmatrix} 1/50 & 1/50 \\ 7/50 & 49/50 \end{bmatrix}$ E) $\begin{bmatrix} 1/50 & 7/50 \\ -7/50 & 7/50 \end{bmatrix}$ F) $\begin{bmatrix} 1/50 & 0 \\ 0 & 7/50 \end{bmatrix}$

2. (12 points) Let $A = \begin{bmatrix} 1 & 1 & 2 & 0 \\ 2 & 4 & 2 & 4 \\ 0 & -1 & 1 & -2 \end{bmatrix}$. Find a bases for $\text{row}(A)$, $\text{col}(A)$ and $\text{null}(A)$.

3. (10 points) Consider the function $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined as $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 5x - y \\ 2x + 3y \\ 5y \end{bmatrix}$. Prove that T is a linear transformation and find the standard matrix of T .